

CSCE 679 DATA VISUALIZATION · MODULE 05 · LECTURE NOTES

Chart Types

Which follow from the earlier modules rather than preceding them.

MODULE OUTCOMES

- Choose a chart from the data and the task.
- Explain distributions and why histograms mislead.
- Explain the specific chart controversies with evidence.
- Explain aspect ratio and axis conventions.
- Justify a chart choice from the ranking.

1 The choice

1. COUNT keys and values (Module 03 section 2), which gives the chart family.
2. NAME the reader's task (Module 01 section 2), which says which variable must be read accurately.
3. ASSIGN that variable to position (Module 02 section 1), and work down the ranking for the others.
4. CHECK that the data types permit the channels you assigned (Module 03 section 1).
5. CHECK the colour (Module 04 section 4).

AND THE CHART TYPE IS WHATEVER THAT PRODUCED.

WHICH IS THE POINT OF THE ORDERING: by this module the choice is nearly determined, and the only remaining decisions are the ones in sections 2 to 4.

WHICH IS WHY THERE IS NO CHART-TYPE DECISION TREE WORTH MEMORISING

A decision tree running from data shape to chart name encodes somebody else's answers to Modules 01 through 04 without showing the reasoning — so it works on the cases they considered and fails silently on yours, with no indication that it has done so.

And it cannot produce a chart nobody has named, which real data frequently requires (Module 01 §1's mark-and-channel framing exists precisely to make that possible).

So the five steps above are the method, and the chart-type galleries are a source of ideas rather than of answers.

Which is worth saying because the galleries are genuinely excellent and are used wrongly — as a menu to choose from rather than as a collection of encodings to learn from, and the difference determines whether you can handle a case they do not contain.

2 Distributions

- A **histogram** — and the **bin width changes the shape**, sometimes dramatically: the same data can appear unimodal or bimodal depending on it, **which makes the bin choice an undisclosed analytical decision** (Module 03 §4's binning point).
- A **kernel density estimate** — which replaces the bin width with a bandwidth and **has exactly the same problem, more smoothly** and therefore less visibly, since the result always looks plausible.
- A **box plot** — compact, and genuinely good for comparing many groups — and **it hides multimodality entirely**: see the callout.
- A **violin plot or a beeswarm** — which shows the actual shape, and **is the right default for a moderate number of points** and a moderate number of groups.
- **And the points themselves, jittered**, which is **the most honest representation and only works below a few hundred points per group**. **Show the points where you can** — every summary above discards the thing you were looking for, which is the shape of the distribution.

A BOX PLOT HIDES EXACTLY THE STRUCTURE YOU WERE LOOKING FOR

A **box plot shows five numbers** — the median, the quartiles, and the whisker extent — and **bimodality, clustering, gaps, and unequal sample sizes are all completely invisible in it**.

So **two groups with identical box plots can have entirely different distributions** — one unimodal and one split into two well-separated clusters — **and the figure asserts that they are the same**.

Which is **Module 01 §3's identical-statistics problem arriving in the chart most commonly used to compare groups**, and is therefore worth knowing specifically rather than as a general caution.

So: **overlay the individual points on the box** wherever the sample size permits — **which costs nothing and restores exactly what the summary removed** — and **state the n per group, which a box plot never shows** and which determines how much any of it means.

3 The controversies

Chart	The objection	The honest position
Pie chart	It encodes on angle and area, ranks four and five (Module 02 §1).	Poor for comparing slices; acceptable for a part-of-whole judgement with very few slices.
Stacked bar	Only the bottom series has a common baseline.	Fine for reading the total and poor for comparing the middle series — so it depends entirely on the task.
Dual axis	The apparent crossing point is arbitrary.	Genuinely misleading; use two stacked panels sharing an x axis — see the note.
3D bar chart	Occlusion, foreshortening, and a false depth channel.	No defence; it is strictly worse than the 2D version in every respect.
Radar / spider chart	Area scales as the square of the values, and the axis order changes the shape.	Hard to read; a parallel coordinates plot or a small-multiple bar chart does the job better.

The dual axis is the one that actually deceives — the apparent crossing or divergence of the two series is a function of the two independent scalings, which the author chose and the reader cannot see — so any desired relationship can be manufactured by adjusting one axis. **Distinguishing the merely weak charts from the actively misleading ones is worth doing**; a pie chart is a poor encoding and an honest one, and a dual axis is a specific deception (Module 10 §2).

AND THE DATA-INK ARGUMENT DID NOT SURVIVE MEASUREMENT INTACT

The claim was that non-data elements should be minimised — gridlines, frames, redundant labels, decorative imagery — and that 'chart junk' actively harms reading, stated forcefully and very influentially.

And the direction is right for genuine clutter: heavy gridlines, 3D effects, drop shadows, and background images do measurably interfere with reading, which the studies confirm.

But the stronger version did not replicate: some redundancy aids reading, gridlines measurably aid value estimation, and memorable decoration improved recall in controlled studies without harming accuracy — so the minimisation principle was overstated.

So the honest position is: remove what interferes, and keep what helps a stated task — which is considerably less quotable and considerably more defensible, and is a good illustration of why this course states mechanisms rather than rules (CSCE 671 Module 05 §1): a rule with no mechanism cannot be corrected by evidence.

4 Axes and aspect ratio

- **A bar chart's value axis must include zero**, because **length is read as a ratio from the baseline** (Module 02 §1's shared-reference argument) — **so truncating it misstates every comparison in the figure**, by an amount the reader cannot recover.
- **A line chart's axis need not include zero**, because **position is read against the scale rather than against a baseline** — **which is precisely why the two conventions differ**, and is the reason rather than an arbitrary distinction.
- **And the aspect ratio changes the perceived trend** — the same series looks flat in a wide figure and steep in a tall one — **which makes it a reportable choice** rather than a layout detail.
- **Banking to 45 degrees is the classical answer: slopes are judged most accurately when they are near 45 degrees**, so choose the aspect ratio to put the trend of interest near that angle.
- **And a log axis must be labelled as such prominently**, because **readers extrapolate linearly from a log plot if they do not notice the scale** — which produces order-of-magnitude errors in the conclusion. **The bar-versus-line zero distinction is the one people get wrong in both directions, and the reason is the channel rather than a convention**, which is what makes it memorable.

WHICH IS THE MODULE'S SUMMARY

Every rule in this module is a consequence of the channel ranking or of the data types — including the ones that are usually stated as bare conventions, like the zero-baseline rule and the pie chart objection.

Which means you can derive them rather than remembering them, and more importantly can decide a case the conventions do not cover — which is what the five-step method in §1 is for.

And where the evidence is mixed, say so — the data-ink case shows that confident design advice can be partly wrong, and the mechanism is what survives when the rule does not.

So: derive the choice, state the task it serves, and test it on readers (Module 13) — which is this course's whole method in one line and is what Project 1 asks you to demonstrate.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Wilke, the directory of visualisations and the pitfalls chapters (free)	§2 through §4 — the distribution charts and the axis conventions, with examples of each failure.
Tufte — The Visual Display of Quantitative Information	§3's data-ink argument in the original — read it alongside the replication work below. Library copy.
Bateman et al. — Useful Junk? (free)	§3's callout, measured — the study that found decoration aided recall without harming accuracy.

Resource	What it covers for this module
Cleveland — The Elements of Graphing Data	§4's banking argument and the aspect ratio material — where it was developed. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Run the five-step choice** on three datasets and record what chart it produced.
- 2 **Find a case the galleries do not cover** and design for it from the steps.
- 3 **Plot the same data at four bin widths** and report how the shape changes.
- 4 **Construct two distributions with identical box plots** and different shapes.
- 5 **Overlay the points** and confirm the difference becomes visible.
- 6 **Find a dual-axis chart** and rescale one axis to reverse the apparent conclusion.
- 7 **Replace it with two panels** and compare.
- 8 **Truncate a bar axis and a line axis**, and explain why only one is a deception.
- 9 **Vary the aspect ratio** of a time series and ask five people whether the trend is strong.
- 10 **Find a log axis that is not prominently labelled.**

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the five-step chart choice.
- 2 Why is there no decision tree worth memorising?
- 3 Name five ways to show a distribution, in increasing honesty.
- 4 What does a histogram's bin width do, and why is a density estimate no better?
- 5 What does a box plot hide, and what is the fix?
- 6 Name five contested charts and the honest position on each.
- 7 Which one actually deceives, and why?
- 8 What survived of the data-ink argument and what did not?
- 9 Why must a bar axis include zero and a line axis need not?
- 10 What does aspect ratio do, and what is banking to 45 degrees?